

Worksheet: Sine and Cosine

Right triangles, the unit circle, and graphs

Name: _____ Date: _____

Goals of this worksheet

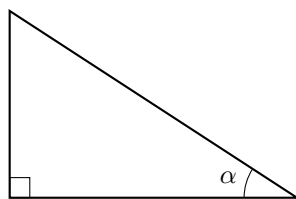
After completing this worksheet, you should be able to

- identify the hypotenuse, opposite side, and adjacent side in a right triangle,
- use the definitions of sine, cosine, and tangent,
- connect sine and cosine to coordinates on the unit circle,
- read and sketch the graphs of sine and cosine.

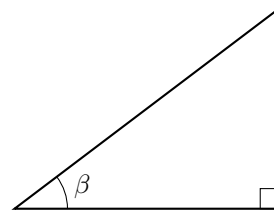
Part A: Right Triangles

Exercise 1: Name the sides

For each triangle, mark the **hypotenuse** with H , the side **opposite** the marked angle with O , and the **adjacent** side with A .



Triangle 1



Triangle 2

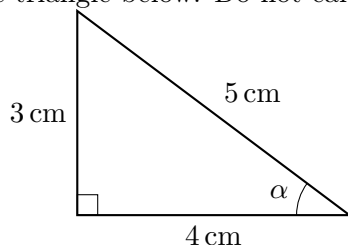
Exercise 2: Complete the definitions

Fill in the missing words.

$$\sin(\alpha) = \frac{\text{opposite}}{\text{hypotenuse}} \quad \cos(\alpha) = \frac{\text{adjacent}}{\text{hypotenuse}} \quad \tan(\alpha) = \frac{\text{opposite}}{\text{adjacent}}$$

Exercise 3: Calculate ratios

Use the triangle below. Do not calculate the angle itself.



a) $\sin(\alpha) = \frac{3}{5}$

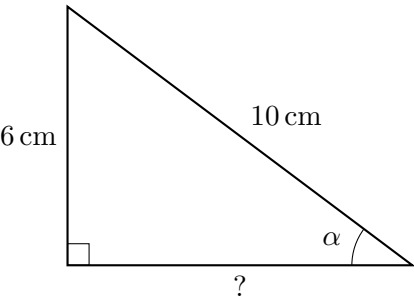
b) $\cos(\alpha) = \frac{4}{5}$

c) $\tan(\alpha) = \frac{3}{4}$

d) Which ratio is the largest? $\frac{3}{4}$

Part A continued: Using Trigonometric Ratios

Exercise 4: Find the missing side
A right triangle has a hypotenuse of 10 cm. The side opposite the angle α has length 6 cm.



a) Use Pythagoras' theorem to calculate the adjacent side. Show your steps.

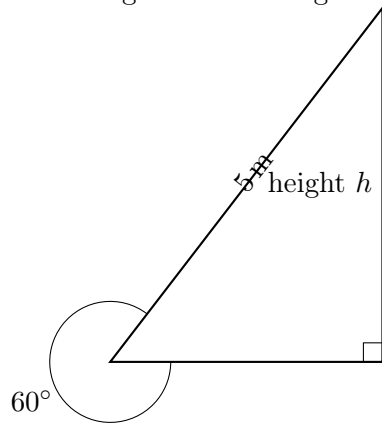
b) Calculate $\sin(\alpha)$ and $\cos(\alpha)$.

Exercise 5: Which trigonometric function?
For each situation, choose sin, cos, or tan. You do not have to calculate anything.

Known and wanted sides	sin	cos	tan
Opposite and hypotenuse	☐	☐	☐
Adjacent and hypotenuse	☐	☐	☐
Opposite and adjacent	☐	☐	☐

Exercise 6: A ladder

A ladder of length 5 m leans against a wall. The angle between the ladder and the ground is 60° .



- a) Which trigonometric function connects h , 5 m, and 60° ?

- b) Write down an equation for h .

- c) Calculate the height. Use $\sin(60^\circ) \approx 0.866$.

Part B: Degrees, Radians, and the Unit Circle

Exercise 7: Degrees and radians

Complete the table. Remember that $360^\circ = 2\pi$.

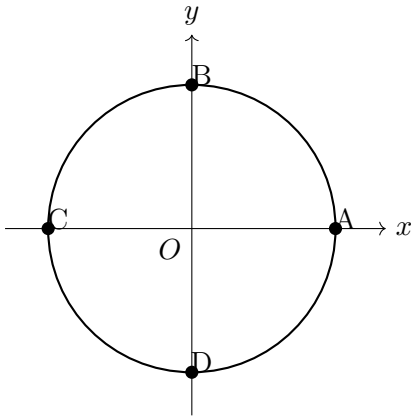
Degrees	Radians	Degrees	Radians
0°		180°	
30°		270°	
45°		360°	
60°			$\frac{3\pi}{2}$
90°			$\frac{\pi}{4}$

Exercise 8: Coordinates on the unit circle

The radius of the unit circle is 1. For an angle α , the point on the circle has coordinates

$(\cos \alpha, \sin \alpha)$.

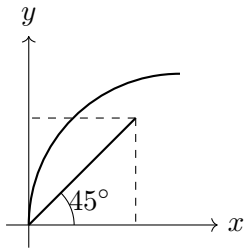
Label the four marked points with their coordinates and complete the table.



Angle	$\cos \alpha$	$\sin \alpha$
0		
$\frac{\pi}{2}$		
π		
$\frac{3\pi}{2}$		
2π		

Exercise 9: A special angle

The point corresponding to 45° lies on the line $y = x$. Therefore, its two coordinates are equal.



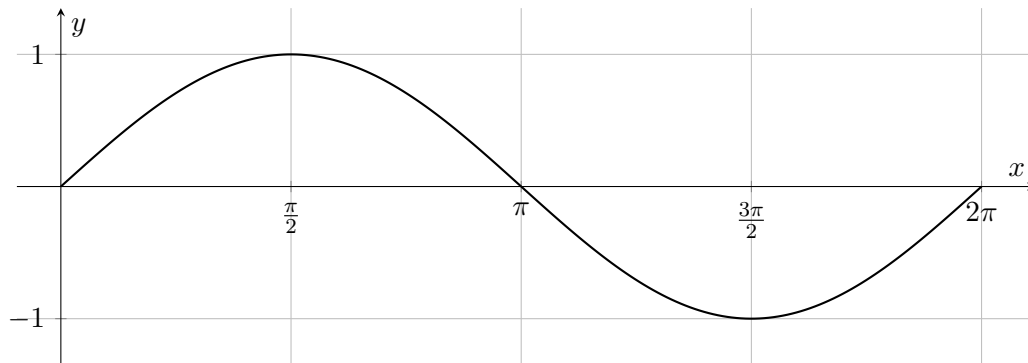
Using $x^2 + y^2 = 1$ and $x = y$, show that

$$\sin(45^\circ) = \cos(45^\circ) = \frac{\sqrt{2}}{2}.$$

Part C: Reading Sine and Cosine Graphs

Exercise 10: Read the sine graph

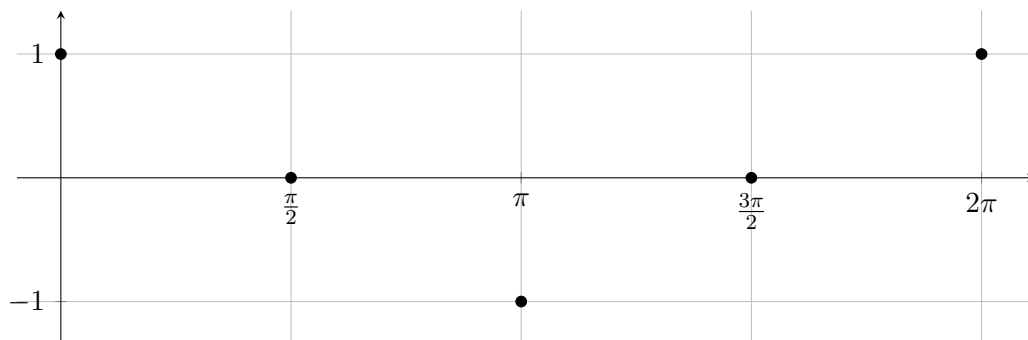
Study the graph and answer the questions below.



- | | |
|---|---|
| a) What is the maximum value? _____ | d) What is the period? _____ |
| b) What is the minimum value? _____ | e) At which x -value is the maximum reached?
_____ |
| c) Where does the graph cross the x -axis?
_____ | f) At which x -value is the minimum reached?
_____ |

Exercise 11: Complete the cosine graph

The marked points belong to $y = \cos(x)$. Connect them with a smooth curve.

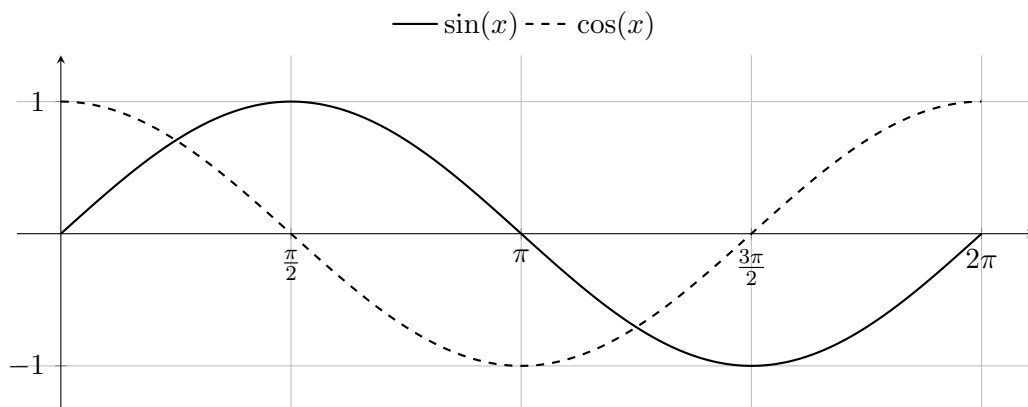


What is the main difference between the starting points of sine and cosine?

Part D: Comparing and Transforming Graphs

Exercise 12: Sine and cosine in one coordinate system

The solid curve is $y = \sin(x)$, and the dashed curve is $y = \cos(x)$.



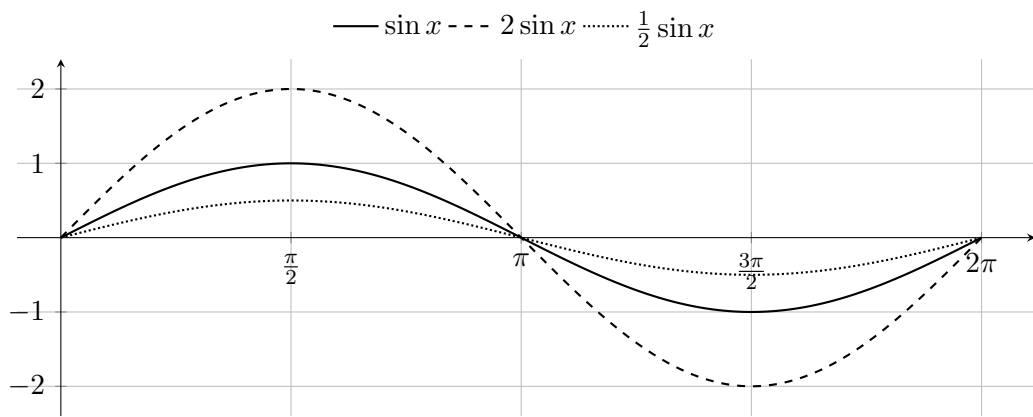
a) What properties do both graphs have in common?

b) By how much must the sine graph be shifted to obtain the cosine graph?

c) Complete: $\cos(x) = \sin(x + \text{_____})$.

Exercise 13: What does the factor in front do?

Compare the three graphs.



a) Complete the table.

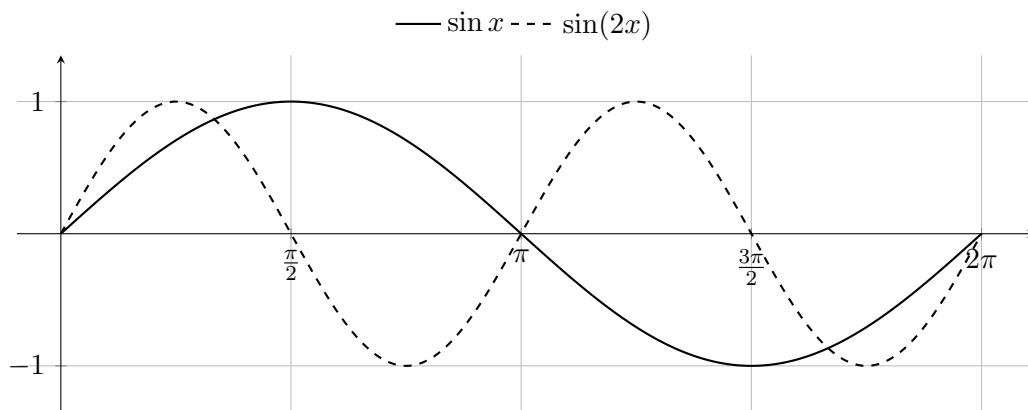
Function	Amplitude	Period
$\sin x$		
$2 \sin x$		
$\frac{1}{2} \sin x$		

What changes when the factor in front of the sine changes?

Part D continued: Frequency and Phase Shift

Exercise 14: What does the factor inside do?

Compare $y = \sin(x)$ and $y = \sin(2x)$.



a) How many complete oscillations does each graph make between 0 and 2π ?

b) Which graph has the shorter period? _____

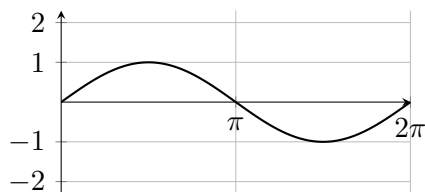
c) Complete: The period of $\sin(2x)$ is _____.

d) Does the amplitude change? _____

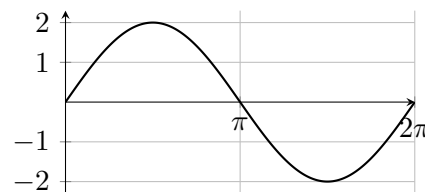
Exercise 15: Match equations and graphs

Match each graph A–D to one equation. Write the correct letter next to each equation.

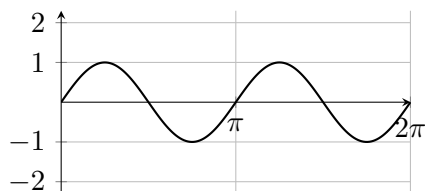
Graph A



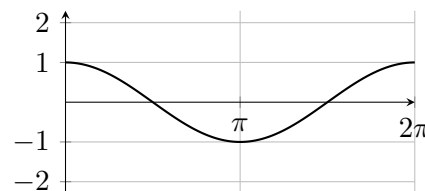
Graph B



Graph C



Graph D



• _____ $y = \sin(x)$

• _____ $y = 2\sin(x)$

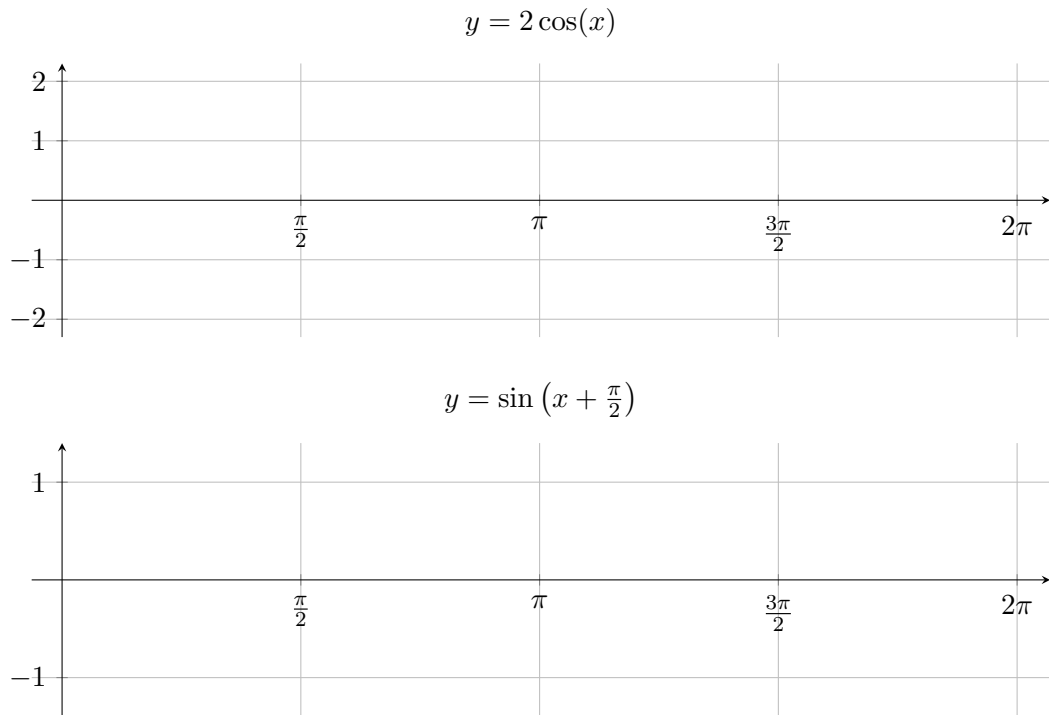
• _____ $y = \sin(2x)$

• _____ $y = \cos(x)$

Part E: Sketching and Explaining

Exercise 16: Sketch the graphs

Sketch one complete period of each function. Mark the zeros, maxima, and minima.



Exercise 17: True or false?

Tick the correct box. Correct every false statement.

Statement	True	False
The sine function has period 2π .	☐	☐
The cosine function starts at 0.	☐	☐
The maximum value of $3 \sin(x)$ is 3.	☐	☐
The function $\sin(2x)$ has amplitude 2.	☐	☐
Sine and cosine have the same shape but are shifted horizontally.	☐	☐

Corrections:

Challenge: Describe a transformed sine function

Consider

$$f(x) = 3 \sin\left(\frac{x}{2} - \pi\right) + 1.$$

Without drawing the graph, determine:

Amplitude: _____ Period: _____

Horizontal shift: _____ Vertical shift: _____

Explain in one or two sentences how the graph differs from $y = \sin(x)$.

Remember: sine and cosine are not only triangle ratios. Their graphs are the basic mathematical shapes of oscillations and waves.